

Tree-Based Methods

Course of Statistical Learning

*Gareth et al.. An Introduction to Statistical Learning,
Ch. 8 and 5.2*

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Slides realized by Alessandro Farinelli and Alberto Castellini

Presentation Outline

Tree-Based
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Tree-Based Methods

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- **Tree-Based** methods for **regression** and **classification**
- **Stratifying** and **Segmenting** predictor space in **simple regions**
 - Splitting Rules
 - Prediction for a given observation: **representative value** for the region of the observation in the training set
 - Splitting rules can be represented as a **tree**
- **Decision Tree** methods
 - Simple and Useful for **interpretation**
 - Usually **not competitive** for prediction accuracy
 - Can combine trees: **bagging, random forest, boosting**
 - **Trade-off** between prediction accuracy and interpretation

Trade-off between prediction accuracy and model interpretability

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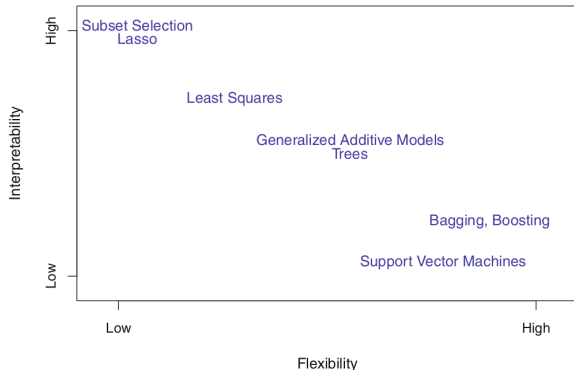
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Graphical representation of flexibility versus interpretability for different statistical learning methods (source [ISL])

Basics of Decision Trees

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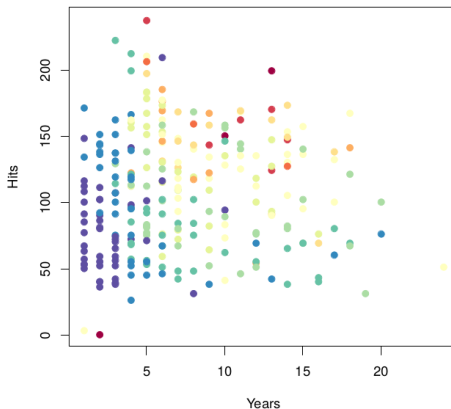
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Visualization of the hitters dataset, source [ISL]



Example of a decision tree

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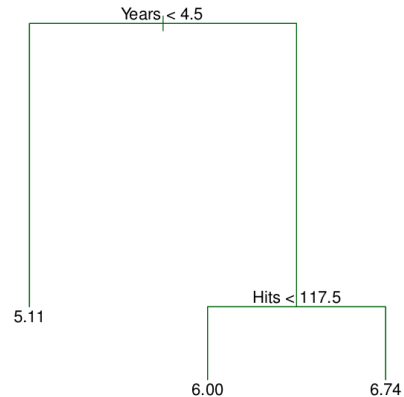
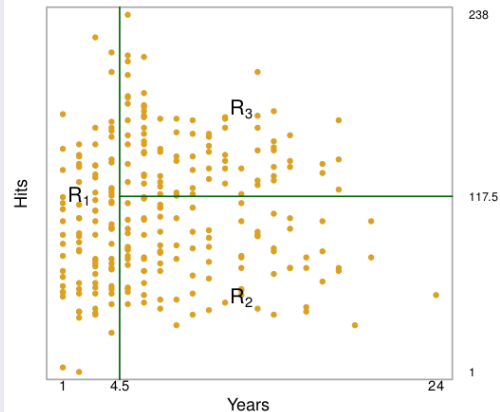
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Regions and decision tree for the hitters dataset, source [ISL]



Basic Terminology

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- **Regions** are called **terminal** nodes
- Representation: **root** at top, **leaves** down
- **Split points** are called **internal** nodes
- Internal nodes are labeled with the **splitting condition**
 - e.g., $Years < 4.5$ and $Hits < 117.5$ are labels for the two internal nodes

Interpretation of results

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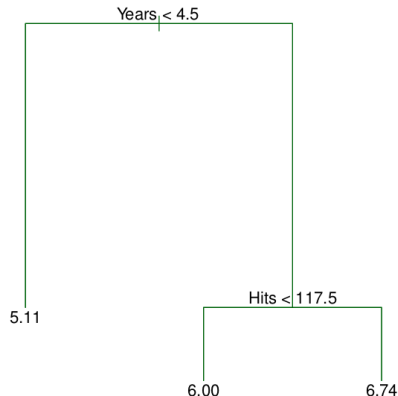
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- *Years* **most important factor** to determine *Salary*
- *Hits* for less experienced players play a **minor** role for *Salary*
- *Hits* are an **important** factor for more experienced players
- Extremely **compact representation** (over-simplification) but much easier to interpret than regression models



Tree building process

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Idea: divide the predictor space into J **non-overlapping regions**

- Predictors: $\{X_1, X_2, \dots, X_p\}$
- Regions: $\{R_1, R_2, \dots, R_J\}$

For every **observation** that falls into the **region** R_j we make the **same prediction**

- **mean of the response value** of all training observations in R_j

Regions could be of any **shape**, we decide to use **boxes** to facilitate interpretation

Goal: find boxes $\{R_1, R_2, \dots, R_J\}$ that **minimize RSS**

$$\sum_{j=1}^J \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

where $\hat{y}_{R_j}$ is the mean response for the training observations within the j -th box

Tree building process: recursive binary splitting

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- **Issue:** not feasible to consider all possible partitions of feature spaces into J boxes
- Use a **Top-Down, Greedy** approach to build the tree
- **Recursive binary splitting**
 - **Top-down:** start from a single region and perform splits
 - **Binary:** always split in two regions
 - **Greedy:** each split optimizes the current step (no look-ahead)

Recursive binary splitting

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Select the **predictor** X_j and the **cutoff value** s such that the **RSS** for the two regions $R_1(j, s) = \{X|X_j < s\}$ and $R_2(j, s) = \{X|X_j \geq s\}$ is **minimized**

- For **all predictors** X_j
- For **all cutoff points** s
- Compute $RSS(j, s) = \sum_{i: x_i \in R_1(j, s)} (y_i - \hat{y}_{R_1(j, s)})^2 + \sum_{i: x_i \in R_2(j, s)} (y_i - \hat{y}_{R_2(j, s)})^2$ where $\hat{y}_{R_k(j, s)}$ is the **mean response** for the training observations in $R_k(j, s)$
- Select (j^*, s^*) that **minimizes** $RSS(j, s)$

Finding (j^*, s^*) is **fast** specially when p is not too large

Select one region and **repeat the splitting** procedure

Continue until a **stopping criterion** is met (e.g., no region contain more than O observations)

Recursive binary splitting, example

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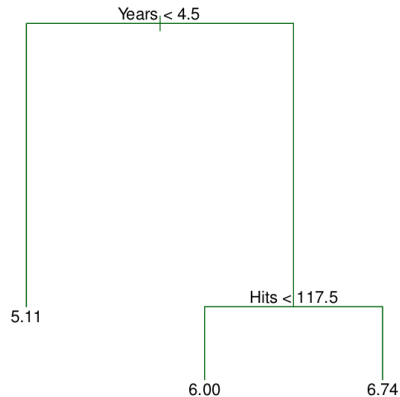
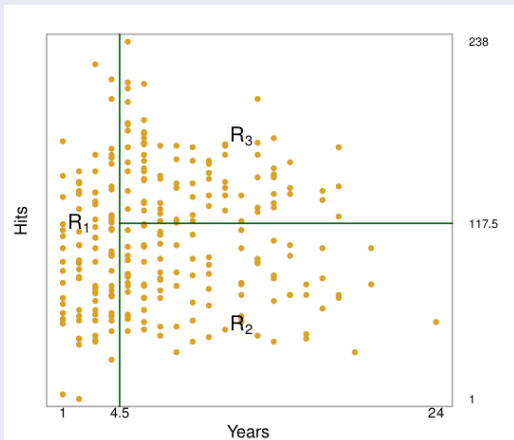
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Recursive binary splitting for the hitters dataset



Predictions

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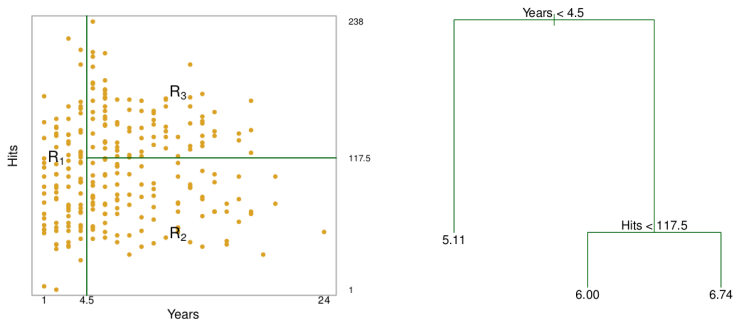
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Compute **response** for a **test** observation:

- determine the **region** to which the test observation belongs
- **return the mean** of the training observations for that region



Example of a decision tree with five nodes

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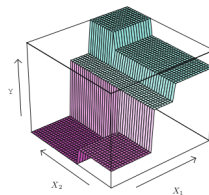
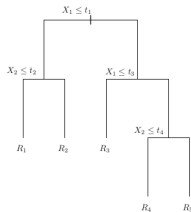
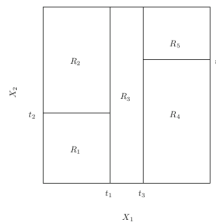
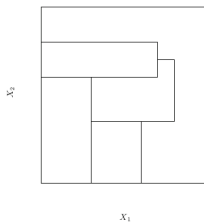
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Pruning the decision tree

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- The **binary recursive splitting** procedure may **overfit**
- **Trade-off** between **variance** (interpretability) and **bias**
- May design a proper **stopping criteria**, e.g., continue to split only if RSS reduction is high
 - This results in **smaller tree**
 - **may have poor results** due to greedy nature of the process: a split that reduces by a small amount the RSS may result in subsequent splits that provide a significant reduction.
- **Idea**: build a very large tree T_0 and then **prune** it back to obtain a **subtree**

Cost-Complexity or weakest link pruning

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- **Cost complexity pruning** (a.k.a., **Weakest link pruning**) is the usual strategy to perform pruning
- Consider a **sequence of trees** indexed by a **parameter** $\alpha \geq 0$
- For each value of α we have a **tree** $T \subset T_0$ that minimizes the following RSS

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

where $|T|$ indicates the **number of terminal nodes of the tree** T

Cost-Complexity or weakest link pruning

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- As α increases, this procedure creates a **series of trees** $T_0 \supset T_1 \supset T_i \supset \dots T_m$ where T_m contains only the root
- Each tree is a **subtree of** T_0 obtained by **collapsing any number of its internal nodes**
- Select best tree based on **cross-validation**
- See <https://youtu.be/D0efHEJsfHo?si=frUeIDCjAYEbfNLG>

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Regression tree procedure

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- 1 **Recursive binary splitting** to build a large tree on training data. **Stopping criterion** on number of observations on terminal nodes (usually small)
- 2 Apply **cost complexity pruning** and obtain a sequence of subtree as a function of α
- 3 Use **K-fold cross-validation** to choose α . For each $k = 1, \dots, K$:
 - 3.1 **repeat steps 1 and 2** of this procedure on all data except the k -th fold
 - 3.2 **evaluate** the mean squared prediction error on the k -th fold as a function of α
- 4 **Average** the results for each value of α and pick the α^* that **minimize** the average error
- 5 **Return the subtree** from step 2 that corresponds to α^*

Example: building a tree for the hitters dataset

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- Randomly divide data in two sets:
 - **Training:** 132 observations
 - **Test:** 131 observations
- Build a large regression tree on the **training** data
- **Prune** the tree using the cost-complexity pruning hence creating a series of subtrees indexed by α
- Perform **6-fold cross-validation** to determine the α^* that minimizes the cross validated MSE

Hitters dataset: large tree

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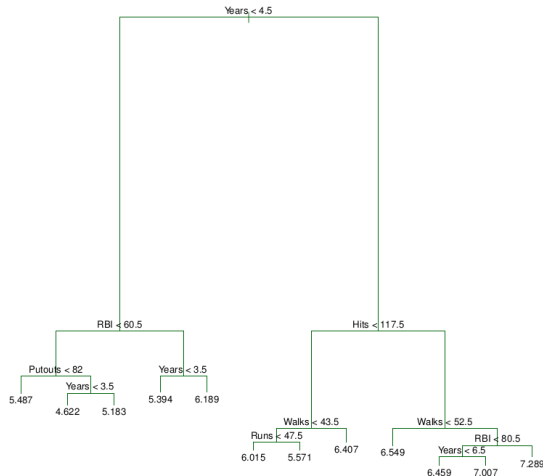
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source [ISL]

Hitters dataset: cross-validation to determine α^*

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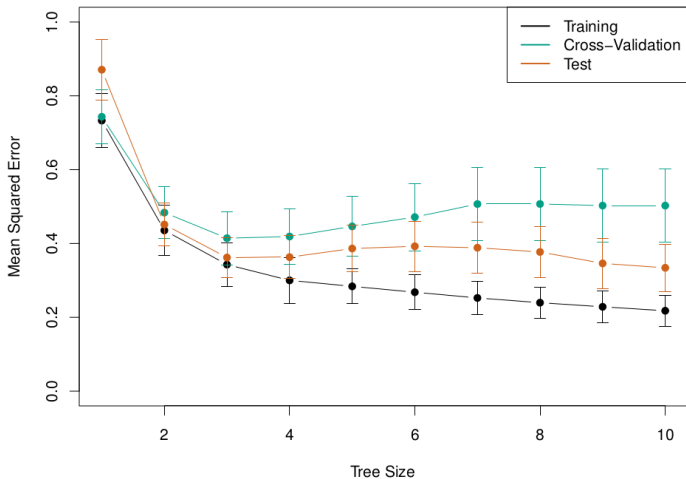
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Classification Trees

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- Similar to regression trees but predict **qualitative** response
- **Prediction: most commonly occurring class of training observations in the region**
- Use **Recursive Binary Splitting** to build the tree
 - Can not use RSS (quantitative not qualitative)
 - Use **classification error rate** instead: $E = 1 - \max_k(\hat{p}_{mk})$
where $\hat{p}_{mk}$ represents the **proportion** of training observations in **region** m that are from the k -th **class**
- Classification error rate is usually **not sensitive enough** for tree-growing
- Use **Gini** index or **Entropy** instead

Gini Index and Entropy

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Gini index:

$$G = \sum_{k=1}^K \hat{p}_{mk}(1 - \hat{p}_{mk})$$

where K is the total number of classes.

- G is **small** if all $\hat{p}_{mk}$ are close to **0 or 1**
- G measures **purity**: it is close to **zero** when a node contains predominantly **observations from a single class**

Entropy:

$$D = - \sum_{k=1}^K \hat{p}_{mk} \log \hat{p}_{mk}$$

G and D are usually very **similar** numerically

Gini Index and Entropy

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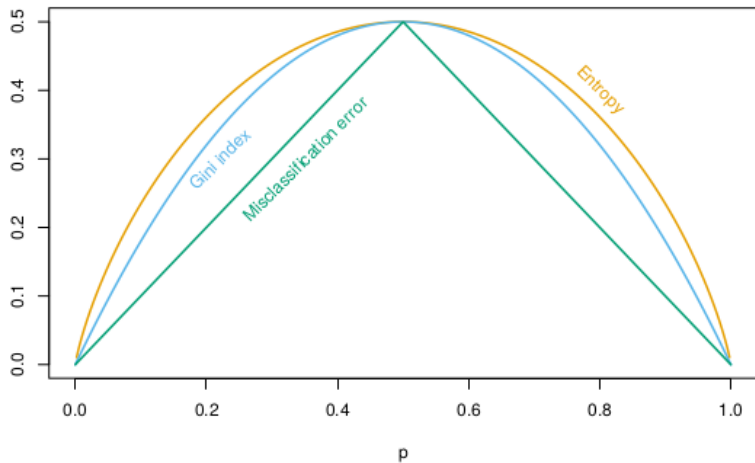
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Example: heart data

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- Contains a binary outcome **HD** for 303 patients suffering from **chest pain**: **Yes/No** means presence/not presence of **heart diseases** (based on angiographic test)
- **13 predictors** including **Age**, **Sex**, **Chol** (cholesterol measurement), as well as other heart and lung function measurements
- **Tree built** using cross-validation has 6 terminal nodes

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Heart dataset: pruned tree and cross-validation error

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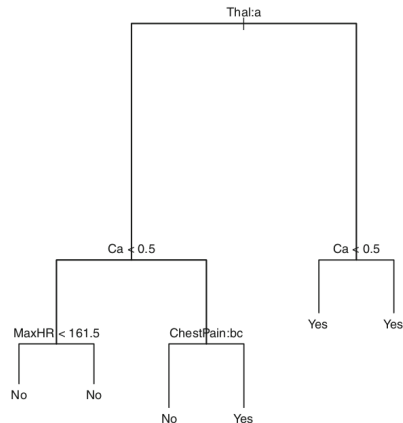
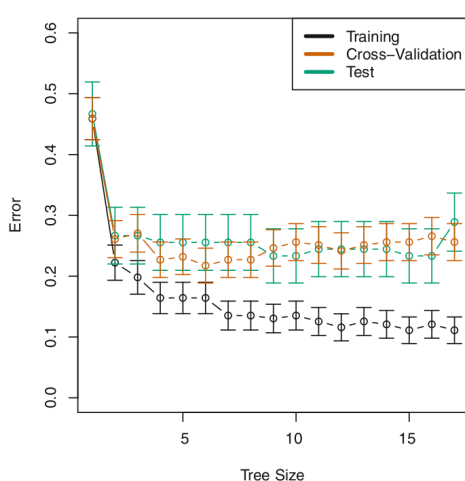
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Linear models vs. tree-based approaches

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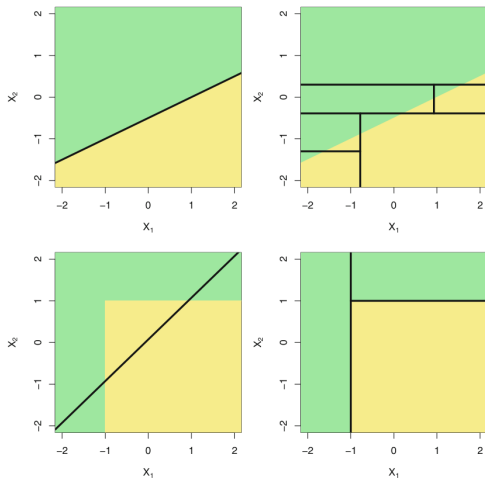
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(source [ISL])

Benefits and Drawback for tree-based methods

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- △ Trees are very **easy to explain** to people
 - △ Trees are very **easy to interpret**
 - △ Great tool to **support decisions**
 - △ Can directly handle **qualitative predictors**
 - ▽ **Not competitive** with other state of the art methods for **prediction accuracy**
-

Predictive accuracy can be significantly increased by **aggregating many decision trees**

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The Bootstrap concept

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- Widely applicable and extremely useful statistical tool
- Can **estimate the uncertainty** of a statistical learning method

Financial example

- Two assets: expected return X and Y
- Expected return of investment: $\alpha X + (1 - \alpha)Y$
- Goal to minimize variance of return: $\text{Var}(\alpha X + (1 - \alpha)Y)$
- Possible to show that $\hat{\alpha}^* = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$

Bootstrap: financial example, computing α

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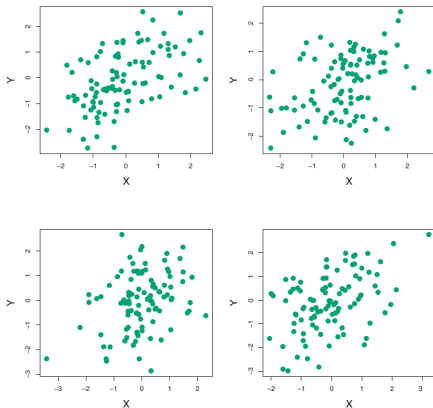
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simulated data set for return of investment (source [ISL])

- 100 simulated pairs (X, Y) on each panel
- estimate $\hat{\sigma}_X^2, \hat{\sigma}_Y^2, \hat{\sigma}_{XY}$ based on data
- compute
 $\alpha = 0.576, 0.532, 0.657, 0.651$
- **What is the variance of α , i.e., the error (uncertainty) that we can get using the estimated α ?**
- **We want to estimate such error (i.e., the standard error of $\hat{\alpha}$)**

Bootstrap: financial example, computing $Var(\alpha)$

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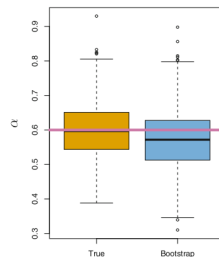
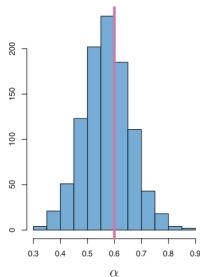
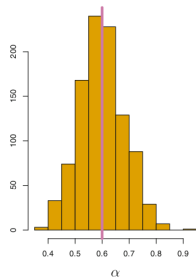
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Estimation of α using more data or bootstrapping (source [ISL])

- left** Generate 1000 **synthetic data sets**, estimate $\hat{\alpha}_r$, $r = 1, 2, \dots, 1000$
($SE(\hat{\alpha}) = 0.086$)
- center** **Bootstrapping**: sample 1000 distinct data sets from a single initial data set
($SE(\hat{\alpha}) = 0.087$, very similar!)
- right** **Comparison** of main statistics for the two estimation of α

Bootstrap: visualization

Tree-Based Methods

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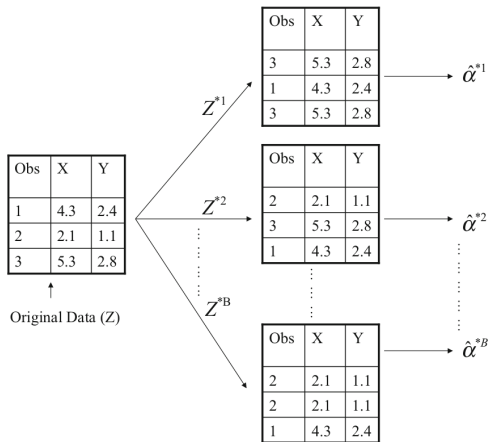
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Visualization of the bootstrap procedure considering a small initial data set Z with $n = 3$ observations (source [ISL])

Bagging I

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- **Bootstrap aggregation**, or **bagging**, is a general-purpose procedure for **reducing variance** of a statistical learning method
- Frequently used in the context of **decision trees**
- Given a set of n **independent observations** $Z_1, Z_2, \dots, Z_n$ each with variance σ^2 , the **variance of the mean** $\bar{Z}$ of the observations is given by σ^2/n
- Averaging a set of observations **reduces the variance**

Bagging II

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- **Bootstrap:** taking **repeated samples** from the (single) training data set
- Generate B different **bootstrapped training sets**
- **Train** a statistical learning method using **each** of the data sets and then **average** the predictions
 - $\hat{f}_{bag}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^{*b}(x)$
where x is a point, $\hat{f}^{*b}(x)$ is the **prediction** for x given by the model trained on the **b -th training set**

Bagging for decision trees

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Regression trees:

- Train B **deep** trees
 - Use the Recursive Binary Splitting procedure
 - Stopping criteria should be tuned to have a deep tree
 - **No pruning**
 - This results in trees with **high variance** but **low bias**
- Given a point x , compute **prediction from each tree** and **average the results**

Classification trees:

- **Train B deep** classification trees, without pruning
- Given a point x , compute the **predicted classes** for each of the B trees and return the **most frequent class** (voting)

Out-Of-Bag error estimation (OOB)

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OOB is a straightforward **way to estimate the test error** for a **bagged model**

- It is possible to show that **on average** each **bagged tree** makes use of about **two-third of the observations**
- **Remaining one third** of observations **not used** to fit a given bagged tree are referred to as the **out-of-bag (OOB)** observations
- We can **predict the response** for the i -th observations using **each of the trees** in which that **observation was OOB**.
- We will have approximatively $B/3$ predictions for each observation for the i -th observation, which we **average**.
- This can be considered as the **LOO cross-validation error for bagging** (when B is large)

Bagging the heart data

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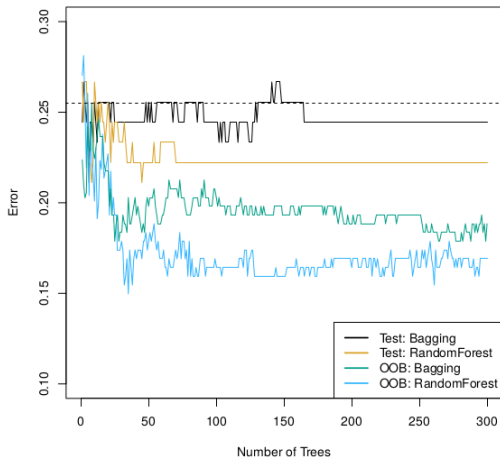
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- **Dashed line** indicates test error for a **single tree**
- Random forest applied with $m = \sqrt{p}$
- OOB: Out Of Bag error

Variable importance measure

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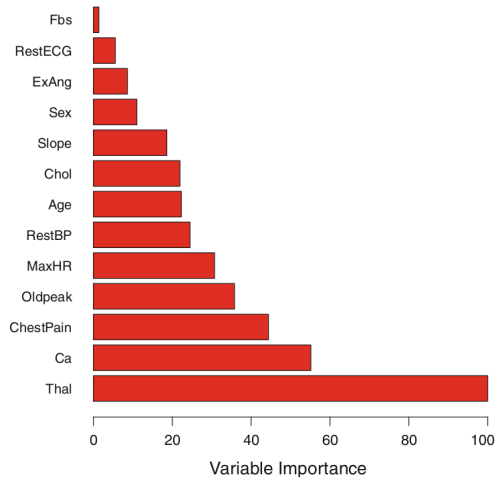
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Variable importance for the heart data set. Variable importance computed by using the mean decrease in the Gini index and expressed relative to the maximum (source [ISL])

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Random Forest

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- Improves upon bagged trees by **de-correlating** the trees
- Similar to bagging we **build a number of decision trees on bootstrapped training samples**
- **Key difference:** when building the decision trees (i.e., RBS):
 - each split is restricted to a **random sample of m predictors** over the p available
 - a **fresh** sample of m predictors is taken at each split
 - typically $m = \sqrt{p}$ (e.g., 4 out of 13 for the heart data set)
- m should be **smaller** if we have a **large** number of **correlated predictors**
- Having **not correlated trees** is crucial to **reduce variance** through bagging (see slides about LOO)

Random forest applied to gene expression data

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Gene expression data

- **High dimensional** biological dataset:
 - 4718 gene expression measurement
 - **349 patients**
 - each patient has a label with **15 possible values**: normal or 14 types of cancer
- Selected **500 genes** that exhibit the largest variance in the training set (data preparation)
- Divided observations into a **training** and a **test** set
- Applied **random forest** with three different values for m

Results: gene expression data set

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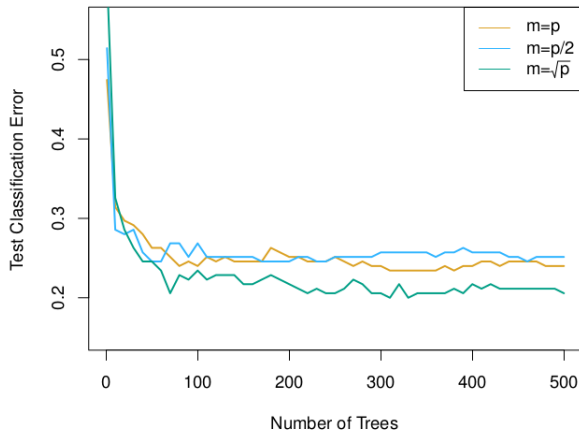
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Results for random forest applied to gene expression data, (source [ISL])

Boosting

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- Another approach for **improving decision trees**
- **General approach** that can be applied to many statistical learning methods
- **Bagging (and random forest)** grows tree on **copies of the original data set** obtained through **bootstrapping**
- **Key observation:** each tree is grown independently from the other trees
- **Boosting: trees are grown sequentially**, each tree is grown using information from previous trees
- Boosting does **not** use **bootstrapping**: each **tree** is fit on a **modified version of the original data set**

Boosting Algorithm for regression trees

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1. Set $\hat{f}(x) = 0$ and $r_i = y_i$ for all i in training set
2. For $b = 1, 2, \dots, B$
 - 2.1 Fit a tree $\hat{f}^b$ with d splits ($d + 1$ terminal nodes) to the training data (X, r)
 - 2.2 Update $\hat{f}$ by adding a shrunk version of the new tree:

$$\hat{f}(x) = \hat{f}(x) + \lambda \hat{f}^b(x)$$

- 2.3 Update the residuals

$$r_i \leftarrow r_i - \lambda \hat{f}^b(x_i)$$

- 3 Output the boosted model

$$\hat{f}(x) = \sum_{b=1}^B \lambda \hat{f}^b(x)$$

Results: gene expression data set

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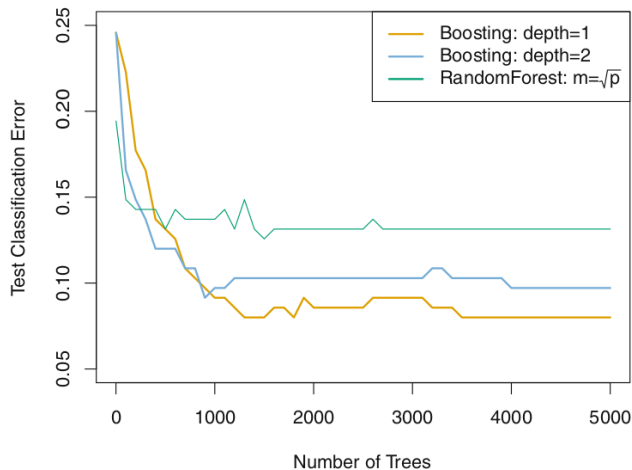
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Results for random forest and boosting applied to gene expression data, $\lambda = 0.01$, (source [ISL])

Tuning parameters for boosting

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1. **Number of trees B** : large B can result in **overfitting**. This overfitting is usually slow. **Cross-validation** to select B
2. **Shrinkage parameter λ** : controls **speed of learning**. Values are usually around 0.01 or 0.001. Related to B , **small** values of λ typically require **large** values of B
3. **Number of splits d** , number of splits in each tree. Often $d = 1$ works well, in this case each tree is a **stump** consisting of a **single split** and resulting in an **additive model**. More generally d is the **interaction depth** and controls the **interaction order** of the boosted model (d **splits** can interact at most d **variables**).

Boosting for classification

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- Boosting for classification is **similar** but somewhat more involved
- We briefly see a very important example of boosting for a specific type of classification trees: **AdaBoost**
 - AdaBoost.M1: Freund and Schapire (1997), a.k.a Discrete AdaBoost
- More details: [The Elements of Statistical Learning](#), chapter 10.
- Current **State of the art** approaches for **classification and regression** are based on **Gradient Boosting**
 - **regularized additive** model
 - second-order approximation to perform optimization
 - evolution of same principle of Ada Boost
 - [XGBoost](#) one of the most **popular** tree boosting system
 - [XGBoost explanation \(video\)](#)

AdaBoost

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- Two class problem $Y \in \{-1, 1\}$, X predictors, classifier $G(x)$

- **Error rate:**

$$\overline{err} = \frac{1}{N} \sum_{i=1}^N I(y_i \neq G(x_i))$$

- **Weak classifier:** error rate slightly better than random guessing
- **Boosting:** sequentially apply weak classification to repeatedly modified versions of the data
- Produce a **series of weak classifiers** $G_m(x)$, where $m = \{1, \dots, M\}$
- **Combine predictions** from weak classifier by a **weighted majority vote**:

$$G(x) = \text{sign} \left(\sum_{m=1}^M \alpha_m G_m(x) \right)$$

- **Weights** $\omega_1, \omega_2, \dots, \omega_N$ are applied to **training observations** (x_i, y_i) to give more **importance** to observations that are **difficult** to classify

AdaBoost.M1 algorithm

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1. **Initialize** observation weights $\omega_i = \frac{1}{N}$, $i = \{1, 2, \dots, N\}$
2. For $m = 1, 2, \dots, M$

2.1 **Fit** a classifier $G_m(x)$ to the training data using weights ω_i

2.2 Compute:

$$err_m = \frac{\sum_{i=1}^N \omega_i I(y_i \neq G_m(x_i))}{\sum_{i=1}^N \omega_i}$$

2.3 Compute $\alpha_m = \log\left(\frac{1-err_m}{err_m}\right)$

2.4 Set $\omega_i \leftarrow \omega_i \exp[-\alpha_m I(y_i \neq G_m(x_i))]$, $i = \{1, 2, \dots, N\}$

- 3 Output $G(x) = \text{sign}\left(\sum_{m=1}^M \alpha_m G_m(x)\right)$

AdaBoost.M1 example: simulated data

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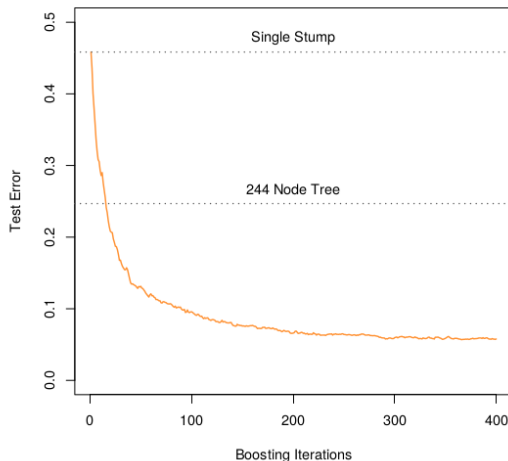
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Results for simulated data(sum of 10 standard Gaussian distributions), weak classifier is a **stump** with 45% test set error rate (source Elements of Statistical Learning, chapter 10).

Summary

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- Decision trees are **simple** and **interpretable** models for regression and classification
- Usually **not competitive** with other methods in terms of prediction accuracy
- Bagging, random forest and boosting can significantly **improve prediction accuracy** of decision trees
 - all of them work by growing **several trees** and then **combining** the predictions of the resulting ensemble of trees
 - **random forest** and **boosting** are among the **state of the art** approaches for supervised learning
 - trade-off between interpretability and prediction accuracy